

This assignment will not be graded and is only for practice.

Key Points: 1

A map $T : V \rightarrow W$ between two vector spaces V and W is said to be a linear transformation if for any two vectors v_1 and v_2 in the vector space V and for any scalar c in $\mathbb{R}$ the following conditions hold:

$$T(v_1 + v_2) = T(v_1) + T(v_2)$$

$$T(cv_1) = cT(v_1)$$

1) Let T be a function from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined as follows:

1 point

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (2x, y)$$

Choose the set of correct option(s).

[Hint: Assume $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2)$, then check the properties of linear transformation for the given function T .]

- ☐ $T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in \mathbb{R}^2$.
- ☐ $T(cv_1) = cT(v_1)$ for all $v_1 \in \mathbb{R}^2$, and $c \in \mathbb{R}$.
- ☐ T is a linear transformation.
- ☐ T is not a linear transformation.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in \mathbb{R}^2$.
 $T(cv_1) = cT(v_1)$ for all $v_1 \in \mathbb{R}^2$, and $c \in \mathbb{R}$.
 T is a linear transformation.

2) Let T be a function from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined as follows:

1 point

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x - 1, y + 1)$$

Choose the set of correct option(s).

[Hint: Assume $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2)$, then check the properties of linear transformation for the given function T .]

- ☐ $T(v_1 + v_2) = T(v_1) + T(v_2)$, for all $v_1, v_2 \in \mathbb{R}^2$.
- ☐ $T(cv_1) = cT(v_1)$ for all $v_1 \in \mathbb{R}^2$, and $c \in \mathbb{R}$.
- ☐ T is a linear transformation.
- ☐ T is not a linear transformation.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is not a linear transformation.

Key Points: 2

A linear transformation $T : V \rightarrow W$ is said to be an isomorphism if it is bijective, i.e. the followings hold:

$T(v_1) = T(v_2)$ implies $v_1 = v_2$ (Injectivity).

For any $w \in W$ there exists $v \in V$ such that $T(v) = w$ (Surjectivity).

3) If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$, then which of **1 point** the following options is true?

[Hints:

Hint 1: Let $v_1 = (x_1, y_1)$, $v_2 = (x_2, y_2)$, and assume that $T(v_1) = T(v_2)$. Then check whether $v_1 = v_2$ or not.

Hint 2: Does there exist any $v \in \mathbb{R}^2$ for which $T(v) = (0, 1)$?]

- ☐ T is both one to one and onto.
- ☐ T is one to one but not onto.
- ☐ T is onto but not one to one.
- ☐ T is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is neither one to one nor onto.

4) If $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation defined by $T(x, y) = (2x + 3y, 5x - y, x + 6y)$. Which of the following options is true? **1 point**

[Hint: Does there exist any $v \in \mathbb{R}^2$ for which $T(v) = (1, 0, 0)$?]

- ☐ T is both one to one and onto.
- ☐ T is one to one but not onto.
- ☐ T is onto but not one to one.
- ☐ T is neither one to one nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is one to one but not onto.

Key Points: 3

Standard Ordered basis of $\mathbb{R}^n$:

$$\{e_i \mid i = 1, 2, \dots, n\},$$

where $e_1 = (1, 0, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, ..., $e_n = (0, 0, \dots, 0, 1)$. In general, $e_i = (0, 0, \dots, 0, 1, 0, \dots, 0)$, where 1 is at the i -th position and 0 elsewhere.

If $T(e_i)$ is known for all i , then the description of T can be known entirely. Suppose $T(e_i) = v_i$, for $i = 1, 2, \dots, n$, then we have

$$\begin{aligned} T(x_1, x_2, \dots, x_n) &= T(x_1 e_1 + x_2 e_2 + \dots + x_n e_n) \\ &= T(x_1 e_1) + T(x_2 e_2) + \dots + T(x_n e_n) \\ &= x_1 T(e_1) + x_2 T(e_2) + \dots + x_n T(e_n) \\ &= x_1 v_1 + x_2 v_2 + \dots + x_n v_n \end{aligned}$$

5) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation, such that $T(1, 0) = (3, 2)$ and $T(0, 1) = (1, 5)$. Which of the following options is true?
[Hint: $\{(1, 0), (0, 1)\}$ is the standard ordered basis of $\mathbb{R}^2$.]

1 point

- ☐ $T(x, y) = (3x + y, 2x + 5y)$
- ☐ $T(x, y) = (3x + 2y, x + 5y)$
- ☐ $T(x, y) = (5x, 6y)$
- ☐ $T(x, y) = (4x, 7y)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(x, y) = (3x + y, 2x + 5y)$$

6) Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear transformation, such that $T(2, 1) = (3, 2)$ and $T(1, 3) = (0, 0)$. Which of the following options is true?
[Hint: Express $(1, 0)$ and $(0, 1)$ in terms of $(2, 1)$ and $(1, 3)$.]

1 point

- ☐ $T(x, y) = \frac{1}{5}(3x, 2y)$
- ☐ $T(x, y) = (x + 1, y + 1)$
- ☐ $T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$
- ☐ T cannot be determined from the given information.

No, the answer is incorrect.

Score: 0

Feedback:

$$\bullet (1, 0) = \frac{3}{5}(2, 1) - \frac{1}{5}(1, 3) \text{ and } (0, 1) = -\frac{1}{5}(2, 1) + \frac{2}{5}(1, 3).$$

$$\text{Hence, } T(1, 0) = \frac{3}{5}T(2, 1) - \frac{1}{5}T(1, 3) = \frac{3}{5}(3, 2) - \frac{1}{5}(0, 0) = \left(\frac{9}{5}, \frac{6}{5}\right)$$

$$T(0, 1) = -\frac{1}{5}T(2, 1) + \frac{2}{5}T(1, 3) = -\frac{1}{5}(3, 2) + \frac{2}{5}(0, 0) = \left(-\frac{3}{5}, -\frac{2}{5}\right)$$

Accepted Answers:

$$T(x, y) = \frac{1}{5}(9x - 3y, 6x - 2y)$$

Key Points: 4

Let $T : V \rightarrow W$ be a linear transformation. Let $\beta = \{v_1, v_2, \dots, v_n\}$ be an ordered basis of V and $\gamma = \{w_1, w_2, \dots, w_m\}$ be an ordered basis of W . So each $f(v_i)$ can be uniquely written as linear combination of w_j 's, where $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

$$T(v_1) = a_{11}w_1 + a_{21}w_2 + \dots + a_{m1}w_m$$

$$T(v_2) = a_{12}w_1 + a_{22}w_2 + \dots + a_{m2}w_m$$

$$\vdots$$

$$T(v_n) = a_{1n}w_1 + a_{2n}w_2 + \dots + a_{mn}w_m$$

The matrix corresponding to the linear transformation T with respect to the ordered bases β and γ is given by,

$$\begin{bmatrix} a_{11} & a_{12} \dots a_{1n} \\ a_{21} & a_{22} \dots a_{2n} \\ \vdots & \vdots \\ a_{m1} & a_{m2} \dots a_{mn} \end{bmatrix}$$

The above matrix not only depends on the linear map T , but also on the choice of the ordered bases β for V and γ for W .

It is important to note that, for a fixed choice of basis of V and W , there is a one to one correspondence between the set of all linear transformations from V to W , and the set of all $m \times n$ matrices, where $n = \dim(V)$ and $m = \dim(W)$.

7) Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x, 0)$. Which **1 point** of the following matrices corresponds to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for both the domain and co-domain?

[Hint: $T(1, 0) = (1, 0)$ and $T(0, 1) = (0, 0)$]

☐ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

8) Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation, such that $T(x, y) = (x + y, x - y)$ **1 point**. Which of the following matrices corresponds to T with respect to the standard ordered basis of $\mathbb{R}^2$, i.e., $\{(1, 0), (0, 1)\}$, for the domain and $\{(1, 1), (1, -1)\}$ for the co-domain?

[Hint: $T(1, 0) = (1, 1)$ and $T(0, 1) = (1, -1)$]

☐ $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Feedback:

• $T(1, 0) = (1, 1) = 1(1, 1) + 0(1, -1)$, and $T(0, 1) = (1, -1) = 0(1, 1) + 1(1, -1)$.

Accepted Answers:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

9) Let $W = \{(x, y, z) \mid x = 2y + z\}$ be a subspace of $\mathbb{R}^3$. Let $\beta = \{(2, 1, 0), (1, 0, 1)\}$ be a basis of W . Let $T : W \rightarrow \mathbb{R}^2$ be a linear transformation such that $T(2, 1, 0) = (1, 0)$ and $T(1, 0, 1) = (0, 1)$. What will be the matrix representation of T with respect to the basis β for W and $\gamma = \{(1, 1), (1, -1)\}$ for $\mathbb{R}^2$? **1 point**

☐ $\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

☐ $\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

☐ $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

10) Let $\beta = \{(1, 0), (0, 1)\}$ and $\gamma = \{(1, 1), (1, -1)\}$ be two bases of $\mathbb{R}^2$. If $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is **1 point**

the matrix representation of a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with respect to β for the both the domain and co-domain. What will be the matrix representation of T with respect to γ for the domain and β for the co-domain?

[Hint: First find $T(x, y)$ and then find $T(1, 1)$ and $T(1, -1)$.]

☐ $\begin{bmatrix} a+b & c+d \\ a-b & c-d \end{bmatrix}$

☐ $\begin{bmatrix} a & -b \\ c & -d \end{bmatrix}$

☐ $\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$

☐ $\begin{bmatrix} a+b & -a-b \\ c+d & -c-d \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Feedback:

• $T(x, y) = (ax+by, cx+dy)$, so $T(1, 1) = (a+b, c+d)$ and $T(1, -1) = (a-b, c-d)$.

Accepted Answers:

$\begin{bmatrix} a+b & a-b \\ c+d & c-d \end{bmatrix}$

Your score is: 0/10